IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

COMMUNICATION NETWORKS

Corrected copy

Tuesday, 12 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : C. Ling

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Special information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,

$1/\mu$ = Average length of packet [bit/packet]

C_i = Transmission speed link i [bits/s]

μC_i = Service rate (link i) [packet/ s]

λ_i = Arrival rate (link i) [packet/ s]

2. Optimal Routing Problem (ORP)

$Min\ D(F)$ with respect to $F = \{ F_i \}$

where, $D(F) = \sum_{i=1}^{L} \frac{F_i}{C_i - F_i}$

and,

C_i = Capacity of link l_i.

F_i = Flow carried by link l_i.

The Questions

1.

a) In the link layer *Go back N ARQ* protocol each frame in error generates K retransmissions.

Assume that you know the probability P of a single frame in error, and that the expected number of transmissions, N_r, of a frame can be approximated by:

$$N_r = \sum_{i=1}^{\infty} f(i) P^{i-1}(1-P)$$

where $f(i) = 1 + (i-1)K$

i) Propose a suitable approximation of K for the case in which $W > 2a+1$ and for $W < 2a+1$. Discuss any assumption made. [5]

ii) Derive the utilisation of the medium for the case in which $W > 2a+1$ and for $W < 2a+1$. [5]

Note: W, is the normalised window size and, a, is the propagation delay / transmission delay ratio.

b)

i)

Describe the main characteristics of mobile ad hoc networks, ensuring that you discuss their main technical issues and the distinct characteristics of their routing algorithms.

[5]

Describe a potential application of a mobile ad hoc network, explaining the specific benefits of using such a network in this application.

[2]

ii) In the context of Intelligent Transportation Systems an intelligent vehicle will be endowed with specialised sensors networks. Briefly discuss three applications that will improve safety and drivers experience.

[3]

2.

a) For the network shown in Fig. 2.1.

i) The social optimal routing problem can be characterised by the cost function:

$$T_1 = \sum_{i \in E} x_i l_i(x_i)$$

Where x_i is the flow in link i and $l_i(x_i)$ is the cost (delay) function of link i.

Discuss what is being optimised when minimising the function T_1. [2]

Solve the social optimal routing, that is, obtain x_1^* and x_2^* that minimise T_1. [2]

ii) The user equilibrium routing problem is characterised by the cost function:

$$T_2 = \sum_{i \in E} \int_0^{x_i} l_i(x_i)$$

Where x_i is the flow in link i and $l_i(x_i)$ is the cost (delay) function of link i.

Discuss why this optimisation problem is also referred as "selfish routing". [2]

Solve the user equilibrium routing problem, that is, obtain x_1^* and x_2^* that minimise T_2. [2]

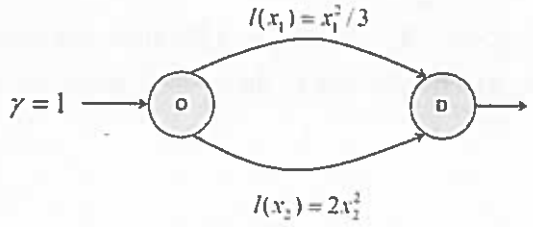


Figure 2.1

iii) Discuss and contrast the physical interpretation of these two solutions. [2]

b)

i) Define and discuss four principles to support quality of service (QoS) guarantees in a TCP/IP network environment. [8]

ii) For each one of the principles identified in i) discuss at least one mechanism that could be implemented in the differentiated service (DS) model. [2]

3.

a)

i) The TCP congestion control algorithm has three major components. Introduce and describe how the three components operate.

[4]

ii) In the context of TCP latency estimation:

Calculate the time it takes the TCP protocol to send an object of size O when $WS/R < RTT + S/R$, stating clearly any assumption made.

Here, W is the size of the window; S is the size of the maximum segment size; R is the transmission rate of the link from server to client; and RTT is the round trip time.

[6]

b) Fig. 1.1 represents a network of M/M/1 queues (links). This network is being fed with a stream of γ_{13} packets/s.

If the aim is to minimise the mean network delay:

i) Find the condition for path 1 (via $C(1)$ and $C(3)$) to start carrying traffic. Assume that initially all the traffic is being carried by path 2 (via $C(2)$).

[5]

ii) If $C(2) = 2\gamma_{13}$ and $C(3) = 2C(2)$ find the minimum capacity $C(1)$ of link 1 to start carrying traffic if initially all the traffic is being carried via link $C(2)$.

[5]

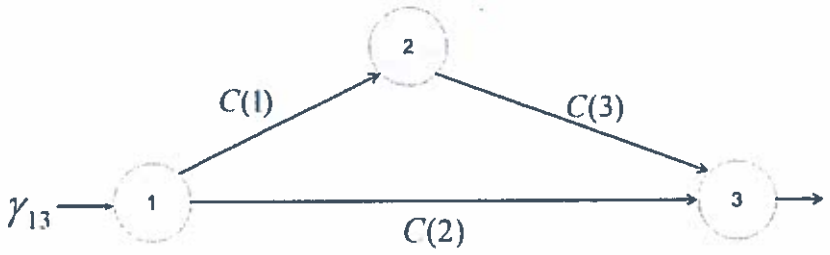


Figure 1.1.

4.

a) The fairness metric in a system that needs to share resources is a measure of how the resources are distributed among the users.

i) The Jain's fairness index is defined as: $\Gamma = \left(\sum_{i=1}^{n} x_i\right)^2 \Big/ \left(n\sum_{i=1}^{n} x_i^2\right)$

Discuss the characteristics of this index. Derive the value of the function Γ in the best and worst case scenario and discuss its physical interpretation. [3]

ii) Define and discuss *max-min fairness* and describe a simple procedure to implement it. [3]

iii) In Fig. 2.2. two applications are represented with their bandwidth utility functions: one elastic traffic application and one adaptive real-time traffic application.

- Identify and sketch the *utility max-min fairness* allocation, [2]
- Identify and sketch the *bandwidth max-min fairness* allocation, [2]
- Discuss the allocation impact on the two contending applications. [2]

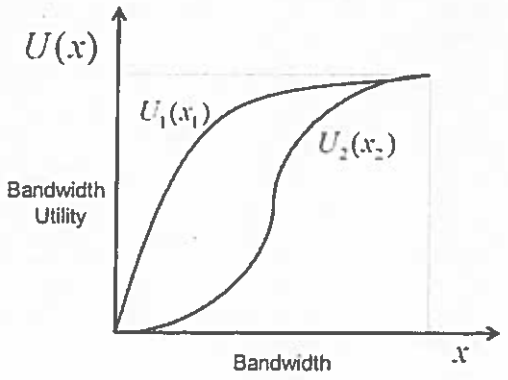


Figure 2.2

b) In the context of a differentiated service (DS) model:

i) Identify and discus the role of the functions that can be deployed at the Edge routers. [4]

ii) Identify and discuss the role of the functions that can be deployed at the Core routers. [4]